

Area Courses

Data Science and Engineering: Core

Fundamental Software Courses

CSCI 455X Introduction to Programming Systems Design 4 units
EE 538 Computing Principles for Electrical Engineers 2 units
EE 547 Applied and Cloud Computing for Electrical Engineers 2 units EE 538

Fundamental Math Courses

EE 503 Probability for Electrical and Computer Engineers 4 units
EE 510 Linear Algebra for Engineering 4 units

Machine Learning

EE 541 A Computational Introduction to Deep Learning 2 units EE 503, EE 510	EE 556 Stochastic Systems and Reinforcement Learning 4 units EE 503 EE 512
EE 559 Machine Learning I: Supervised Methods 3 units EE 541 EE 503, EE 510	EE 641 Deep Learning Systems 2 units EE 541, EE 559
EE 660 Machine Learning II: Mathematical Foundations and Methods 4 units EE 503, EE 510, EE 559 EE 538, EE 541	EE 689 Computational Intelligence and Neural Learning 4 units EE 483, EE 503, EE 510

Mathematical Methods for Data Analytics and Machine Learning

EE 517 Statistics and Data Analysis for Engineers 4 units EE 503	MATH 541ab Introduction to Mathematical Statistics a:3 units, b:3 units
EE 546 Mathematics of High-Dimensional Data 4 units EE 503, EE 510	EE 563 Estimation Theory 3 units EE 503
EE 583 Statistical Signal Processing 3 units EE 503	EE 588 Optimization for the Information and Data Sciences 3 units EE 510 EE 503

Legend

Grouping

EE 000 Course Title

Course Units

Recommended Prep.

Prerequisite Courses

Corequisite Courses

indicates a class with a significant computing/design component

This chart shows course relationships

Please check the University Catalogue for specific course details including any recommended preparatory courses and Degree Requirements